

Variables - Answers

Problem 1: Variables of States

The observational units are the fifty states.

- a) **Not a variable.** This is one total for the entire group of states, so it does not have a different value for each state.
- b) **Variable.** The percentage of residents over age 65 can differ from state to state.
- c) **Variable.** The highest speed limit can differ from state to state.
- d) **Variable.** Each state has a Yes or No value, so this is also a binary categorical variable.
- e) **Variable.** The average income of adult residents can differ from state to state.
- f) **Variable.** The 1999 birth rate is fixed for each state, but it can differ from one state to another.
- g) **Not a variable.** This is one total for the entire group of states, rather than a value recorded separately for each state.
- h) **Not a variable.** The nationwide average number of representatives per state is one summary value for all states.
- i) **Variable.** The number of Congressional representatives can differ from state to state.

Problem 2: Types of Variables

The observational units are the 22 Broadway productions.

- a) **Production: Categorical.** The production name identifies a show.
- b) **Type: Categorical and binary.** The dataset contains two categories: Musical and Play.
- c) **Gross: Quantitative.** Gross measures an amount of money.
- d) **Attendance: Quantitative.** Attendance counts the number of people who attended.
- e) **Capacity: Quantitative.** Capacity records the number of available seats.
- f) **Top price: Quantitative.** Top price measures an amount of money.